

People Analytics

Project Report

at the Faculty of Business, Economics, and Law
Friedrich-Alexander-Universität Erlangen-Nürnberg
Schöller Endowed Chair for Information Systems
Prof. Dr. Sven Laumer

Presented by: Mahad Khalid Tarar
Erlangen
mahad.khalid.tarar@fau.de

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1 Well-Being

1.1 The importance of well-being (4 Points)

Monitoring employee well-being is important because it directly affects both individual and organizational outcomes. Employees with higher well-being tend to perform better, are more engaged, and take fewer sick leaves. Positive well-being, including both mental and physical health, improves morale and supports a healthier, more inclusive workplace culture.

For the organization, high well-being helps reduce health insurance costs, prevent legal issues, and increase productivity. Since well-being is influenced by both work and personal life, companies that actively monitor it can create a supportive environment that enhances overall performance and employee satisfaction.

Low employee well-being and poor work-life balance can result from several interconnected factors. These include poorly designed jobs with little autonomy, long working hours, and lack of social support. Conflicts between work and family commitments, perceived unfairness at work, and economic insecurity (such as fear of layoffs or inadequate pay) also contribute negatively.

Additionally, lack of emotional support, unclear roles, limited career development opportunities, and misalignment between personal values and organizational goals can harm well-being. Poor physical health due to inactivity, unhealthy diets, or unsafe work environments further lowers well-being. When these issues accumulate, they create stress, reduce morale, and lead to burnout.

FAU Transline can monitor employee well-being by adopting a practical and comprehensive approach that goes beyond simply offering formal work-life balance (WLB) policies. Research shows that such programs alone may not significantly improve how employees perceive their health and well-being. Instead, employees' individual strategies for managing work and personal life play a crucial role. To better understand these factors, FAU Transline should regularly collect feedback from its conductors, drivers, and support staff through surveys and open conversations. These should explore how employees handle stress, balance work and family responsibilities, and their unique circumstances such as working hours, job roles, and personal commitments.

To measure well-being in a structured and reliable way, FAU Transline can apply well-established tools. The PERMA Model evaluates key psychological aspects including positive emotions and engagement. The Warwick-Edinburgh Mental Well-Being Scale gathers information about employees' mental health by asking them to report their feelings. The Well-Being Questionnaire assesses physical, emotional, and social health dimensions. Using these validated tools will provide a more detailed understanding of employees' well-being beyond simple satisfaction measures.

Additionally, FAU Transline can implement analytics software to analyze the data collected from these tools. This enables the organization to track well-being trends over time and identify groups or individuals who may require extra support. Combined with regular communication between employees and managers, this approach allows FAU Transline to offer personalized assistance and continuously improve its WLB policies. This well-rounded strategy supports a healthier, more supportive work environment, enabling employees to perform their best both on the job and in their personal lives.

1.2 Stress induced by technology (7 Points)

The planned replacement of FAU Transline's legacy software system with a modern digital platform is a classic example of how information and communication technologies (ICT) can generate technostress. Technostress is defined as stress experienced by individuals when facing challenging or overwhelming situations related to ICT use. This stress can arise from constant exposure to new or changing technology, as well as demands imposed by the digital environment. Given that FAU Transline's employees rely heavily on the current system for daily operations, this transition could significantly impact their stress levels.

According to the Challenge-Hindrance Framework, technostress can manifest in two distinct forms: challenge stressors and hindrance stressors. Challenge stressors are demands perceived as obstacles to overcome, which can motivate employees by offering opportunities for personal growth, skill development, and increased performance. For example, mastering the new system could be seen as a challenge stressor that encourages learning and competence enhancement among conductors and support staff. This positive appraisal can result in improved engagement and productivity.

On the other hand, hindrance stressors are demands that employees view as constraints or threats that prevent growth. These include factors such as system breakdowns, unclear features, and frequent software updates, which can disrupt workflows and cause frustration. Employees might experience techno-overload from increased work demands, techno-complexity from difficulty using the new system, or techno-uncertainty due to continuous changes requiring ongoing learning. Such hindrance stressors tend to demotivate employees and can lead to negative outcomes like reduced performance, job dissatisfaction, and increased burnout risk.

Technostress in the context of FAU Transline can lead to both positive and negative consequences for employees and the organization. On the positive side, when managed well, technostress can enhance employee efficiency and foster greater overall work engagement. It may contribute to a better overall feeling at work, higher individual performance, and encourage innovative use of IT systems. These positive effects can help employees adapt to technological changes and support smoother operations within the company. However, unmanaged technostress can have detrimental impacts. Employees may suffer from reduced job satisfaction, decreased organizational commitment, and lower performance. It can also cause role conflicts and role overload. Additionally, dissatisfaction with the information systems can increase, alongside physical and mental strain such as exhaustion and burnout.

For FAU Transline, these negative consequences can lead to operational inefficiencies, compromised commuter satisfaction, and even safety risks, particularly during critical scheduling and route management tasks.

To cope effectively with this transition, FAU Transline should adopt a dual approach addressing both challenge and hindrance stressors. First, interventions can focus on framing the new platform as an opportunity for growth by providing comprehensive training and encouraging mastery of the system. This would support employees in appraising the change as a challenge, fostering positive coping behaviors and outcomes. Second, the company must minimize hindrance stressors by ensuring system reliability, user-friendly design, clear communication about changes, and timely technical support. Addressing techno-overload and techno-invasion by setting reasonable expectations for after-hours availability is also important.

General recommendations for the company include engaging employees early in the change process, offering continuous learning resources, and providing psychological support to manage strain. Creating a culture where feedback is welcomed can help identify stressors early and allow timely adjustments. Additionally, balancing workload and recognizing employees' efforts can enhance motivation and reduce perceived hindrances. By applying these strategies, FAU Transline can not only reduce technostress but also leverage the digital platform to improve employee satisfaction and operational performance.

In summary, the introduction of a modern ICT platform at FAU Transline presents both opportunities and risks related to technostress. Understanding the dual nature of technostress through the Challenge-Hindrance Framework helps predict potential impacts and guide effective coping mechanisms. A well-planned change management strategy focusing on supportive training, clear communication, and stress mitigation will be essential to ensure a smooth transition and maintain the company's commitment to employee well-being and service excellence.

1.3 Employee well-being dataset (8 Points)

The analysis was carried out using the dataset "wellbeing__of_employees.csv" with the goal of identifying factors that influence employees' Work-Life Balance (WLB) at FAU Transline.

Data Preparation and Cleaning

Before exploring the data, appropriate preprocessing steps were applied to ensure the quality and reliability of results.

First, the column EMPLOYEE_ID was removed as it is only a unique identifier and does not carry any meaningful information for analysis. Next, the TODO_COMPLETED column contained an invalid time-format entry ('14:30') alongside string values. This was handled by converting all values to numeric and dropping the incorrect entry.

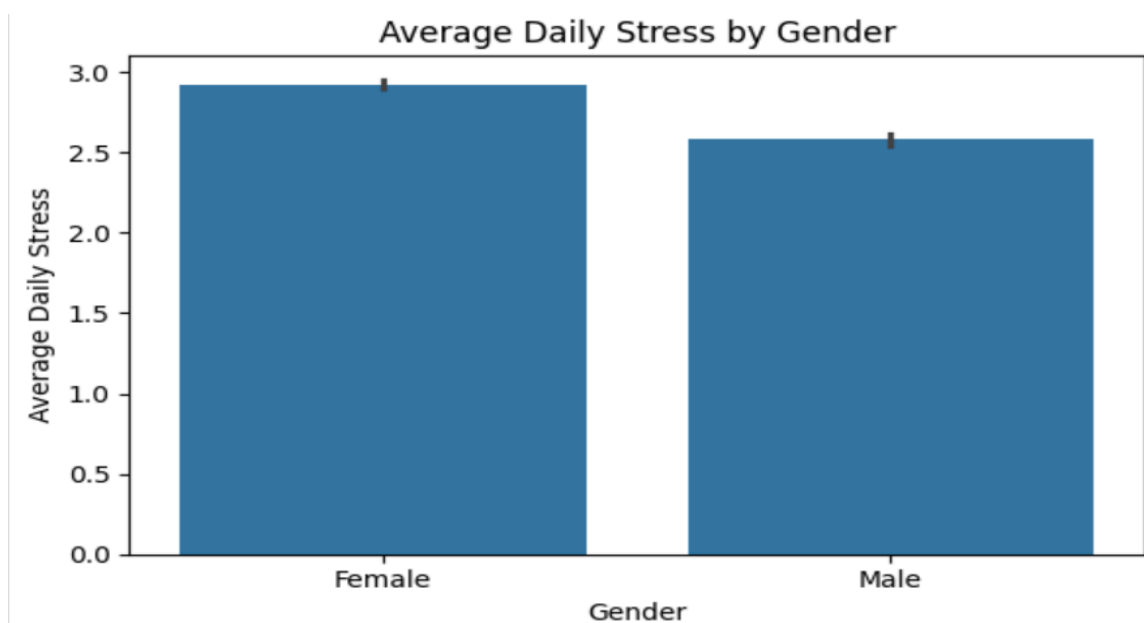
Missing data was minimal; only one row contained a missing value in the DAILY_STRESS column. Since the impact of removing a single row is negligible, it was dropped to maintain consistency in the dataset.

Categorical variables such as GENDER, AGE, and JOB_ROLE were converted into numeric values to make the data suitable for analysis. For example, gender was encoded with 0 representing female and 1 representing male.

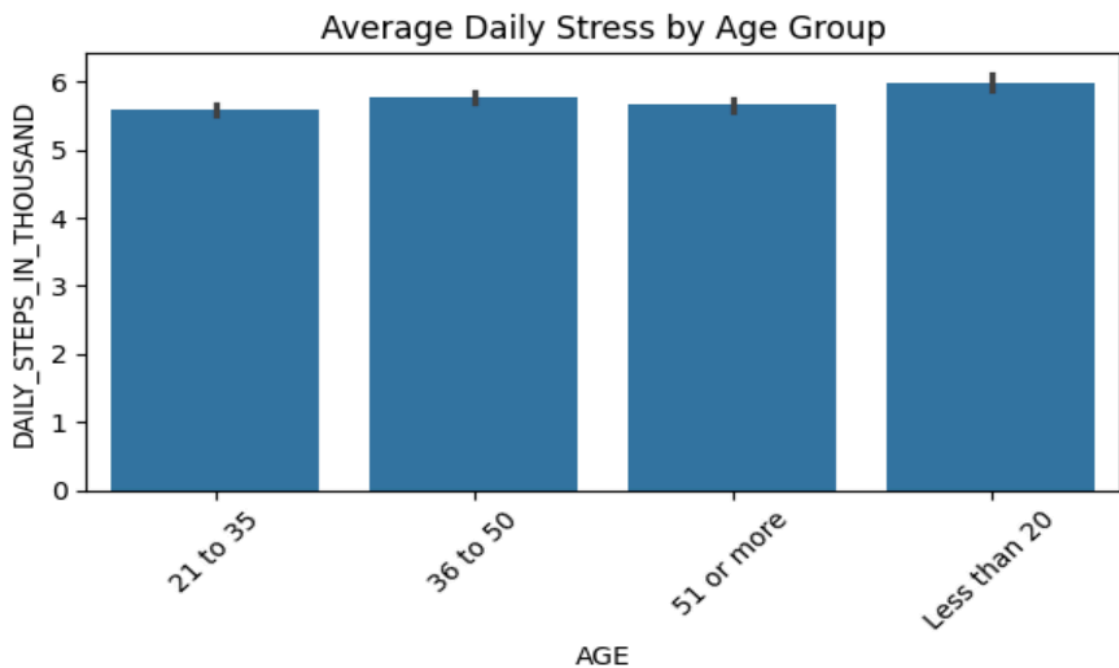
After cleaning and transforming the data, it was ready for further visualization and statistical analysis.

Graphical Representations: Stress and Physical Activity

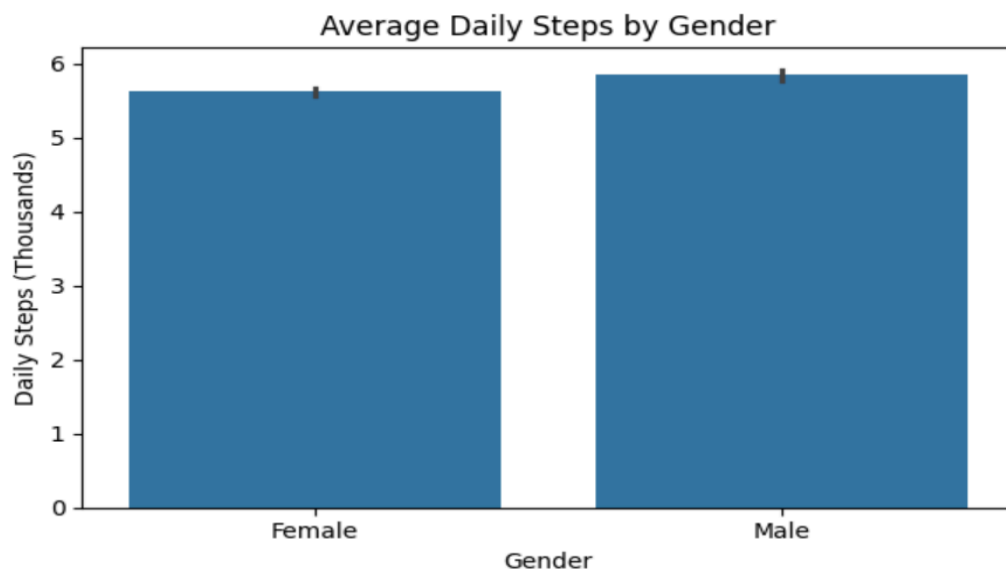
To understand the patterns of stress and physical activity among employees, average values were compared across different demographic groups.



A bar chart of average daily stress by gender shows that female employees tend to experience slightly higher stress levels than male employees.



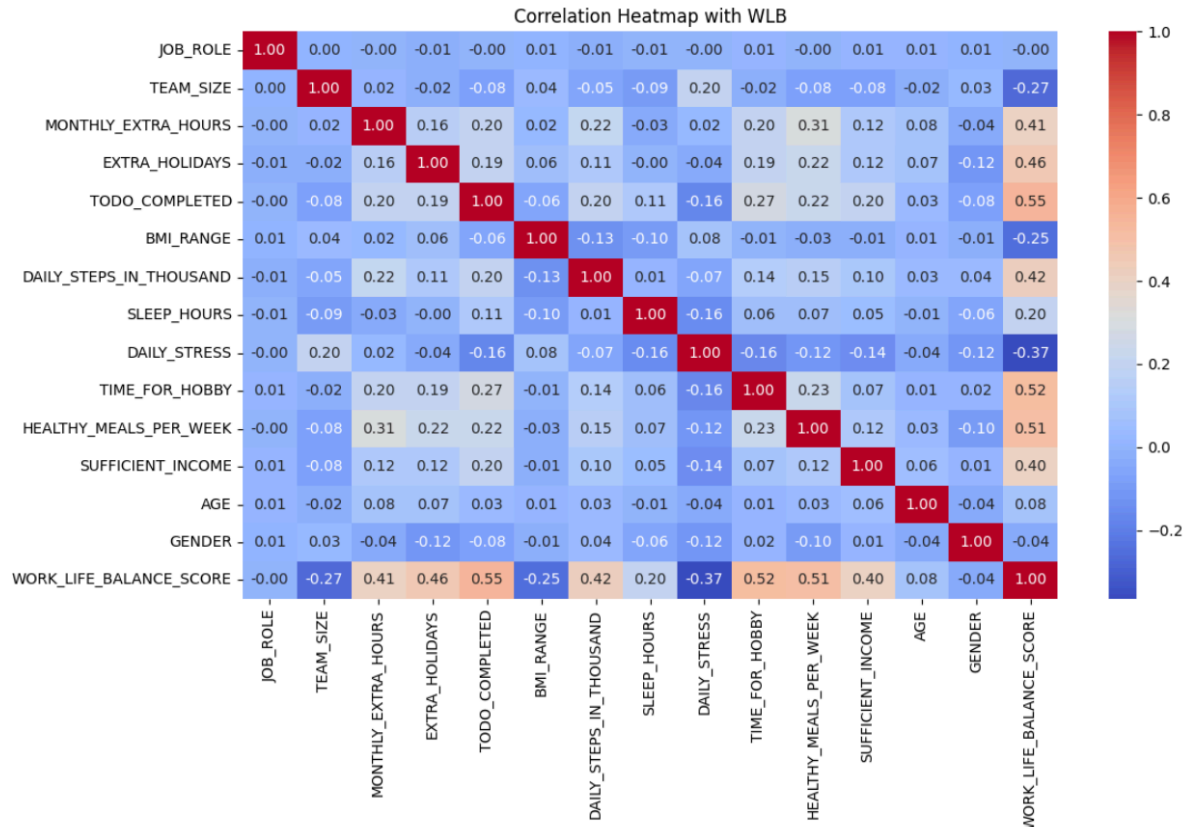
Another comparison based on age groups shows that younger employees, particularly those under 20, report the highest levels of daily stress. Stress levels tend to decrease slightly with increasing age.



In terms of physical activity, average daily steps by gender indicate that male employees take slightly more steps per day than female employees, though the difference is relatively small.

Correlation with Work-Life Balance

To identify the most important factors influencing the WLB score, a correlation analysis was conducted.



The analysis reveals that several factors are strongly and positively associated with higher work-life balance (WLB) scores. Among these, the number of daily tasks completed shows the highest positive correlation at 0.55, indicating that employees who complete more tasks tend to feel a better balance between work and life. Having time for hobbies also correlates strongly (0.52), suggesting that personal leisure activities play a significant role in enhancing WLB. Eating healthy meals regularly (0.51) and receiving extra holidays (0.46) further contribute positively, highlighting the importance of healthy routines and adequate rest. Additionally, taking more daily steps (0.42) and feeling financially secure (0.40) are linked with better WLB, reinforcing the value of physical activity and financial stability in employees' perceptions of balance.

In contrast, certain factors show a clear negative relationship with WLB scores. Higher daily stress is negatively correlated at -0.37, meaning that increased stress levels substantially reduce employees' sense of balance. Larger team sizes (-0.27) and higher BMI values (-0.25) also have a detrimental effect, which may reflect the challenges of managing heavier workloads and poorer physical health on perceived work-life harmony.

Interestingly, variables such as sleep hours have only a weak positive correlation with WLB, and demographic factors like age, gender, and job role show little to no meaningful effect.

This suggests that lifestyle habits and workplace conditions have a more significant impact on employees' experience of work-life balance than inherent personal characteristics. Overall, these findings underscore the importance of promoting healthy daily routines, ensuring time for personal interests, managing stress effectively, and providing adequate compensation and breaks to improve employees' well-being and perception of balance between work and life.

1.4 Predictive well-being algorithm (6 Points)

FAU Transline aims to understand and improve employee work-life balance (WLB) amidst growing operational demands. To support workforce planning and well-being initiatives, a predictive model was developed to estimate the WLB score of employees based on various personal, health, and job-related features.

Model Development

A Linear Regression model was trained using the available dataset, which includes features such as team size, extra working hours, holidays, health indicators, and lifestyle factors. The target variable is the *WORK_LIFE_BALANCE_SCORE* (WLB). Some features with little predictive power (e.g., *JOB_ROLE*) were removed to improve model clarity and performance.

Model Performance

The model achieved a coefficient of determination (R^2 score) of **0.8609**, indicating that approximately 86% of the variability in the WLB scores can be explained by the model. This is a strong result, suggesting good predictive capability.

```
# Define features and label
## Independent Variables
X = df.drop(columns=['WORK_LIFE_BALANCE_SCORE'])
## Dependant Variable
y = df['WORK_LIFE_BALANCE_SCORE']

# Train/test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=0)

# Linear Regression
lr = LinearRegression()
lr.fit(X_train, y_train)
y_pred = lr.predict(X_test)

# R2 Score
r2 = r2_score(y_test, y_pred)
print(f"R2 Score: {r2:.4f}\n")
```

R² Score: 0.8609

Comparison of Actual vs Predicted Values

The following table presents a sample of actual WLB scores from the dataset and their corresponding predicted values by the model for the first 20 employees:

Index	Actual	Predicted
0	688.8	676.18
1	703.4	690.46
2	673.2	676.80
3	662.0	682.51
4	693.7	658.98
5	595.5	591.03
6	649.5	659.69
7	644.4	632.20
8	688.5	703.32
9	660.5	621.24
10	637.3	658.91
11	682.4	667.97
12	620.8	606.17
13	638.6	671.78
14	676.2	668.41
15	674.1	668.06
16	698.1	699.95
17	684.1	685.84
18	702.1	728.68
19	674.7	718.56

Prediction for a New Employee

For demonstration, a new employee record was created with the following assumed attributes:

```
new_employee = pd.DataFrame([{
    'JOB_ROLE': le_job.transform(['Train Conductor'])[0],
    'TEAM_SIZE': 3,
    'MONTHLY_EXTRA_HOURS': 5,
    'EXTRA_HOLIDAYS': 1,
    'TODO_COMPLETED': 7,
    'BMI_RANGE': 2,
    'DAILY_STEPS_IN_THOUSAND': 7,
    'SLEEP_HOURS': 6,
    'DAILY_STRESS': 3,
    'TIME_FOR_HOBBY': 4,
    'HEALTHY_MEALS_PER_WEEK': 6,
    'SUFFICIENT_INCOME': 2,
    'AGE': le_age.transform(['21 to 35'])[0],
    'GENDER': le_gender.transform(['Male'])[0],
}])

new_employee_wlb_prediction = lr.predict(new_employee)
print(f"Predicted WLB score for new employee: {new_employee_wlb_prediction[0]:.2f}")

Predicted WLB score for new employee: 655.26
```

Model Insights from Regression Analysis

The regression analysis provides deeper insights into the factors influencing work-life balance (WLB). Among the positive influences, sufficient income emerges as the strongest contributor to improving WLB, emphasizing the critical role financial security plays in employees' overall sense of balance. Alongside income, having extra holidays, spending time on hobbies, maintaining a healthy diet, getting adequate sleep, and engaging in daily physical activity all significantly enhance work-life balance. The positive correlation between task completion and WLB further highlights the importance of effective workload management, suggesting that employees who can manage and complete their tasks feel more balanced.

On the other hand, certain factors negatively impact WLB. High daily stress levels and unhealthy BMI stand out as major detractors, significantly lowering employees' perception of balance. Larger team sizes also correlate with reduced WLB, likely reflecting the increased coordination difficulties and pressures that come with bigger teams.

Regarding demographics, while age and gender have minor effects, these influences are statistically significant, indicating that although personal characteristics do play a role, their impact is relatively small compared to lifestyle and work-related factors. Overall, the regression results reinforce the notion that financial stability, health, leisure time, and manageable workloads are key to fostering a healthier work-life balance, while stress, poor health, and complex team dynamics can undermine it.

Recommendations for FAU Transline

Based on the findings, the following steps can help improve employee work-life balance:

1. **Enhance Compensation and Benefits:** Ensuring sufficient income is critical. Competitive salaries and financial incentives can directly improve WLB.
2. **Promote Health and Wellness:** Initiatives such as wellness programs, healthy meal options, and fitness activities should be encouraged.
3. **Encourage Work-Life Boundaries:** Support employees in having hobbies and personal time, possibly through flexible scheduling and workload management.
4. **Reduce Stress:** Implement stress management workshops, mental health support, and realistic workload expectations.
5. **Increase Time Off:** Offering more holidays or flexible leave can provide essential recovery time.

6. **Optimize Team Sizes:** Smaller, well-coordinated teams may foster better communication and individual accountability.
7. **Maintain Equity:** Although demographic effects are minor, fair treatment across all employees remains important.

2 Turnover

2.1 Employee Turnover (4 Points)

Voluntary employee turnover refers to an employee's own decision to leave the organization, typically for reasons such as dissatisfaction, better opportunities, or personal preferences. Unlike involuntary turnover, which is initiated by the employer, voluntary turnover reflects an individual's intent to resign and can be classified as avoidable if the organization could have prevented it through better hiring, assessment, or motivational practices.

For a company like FAU Transline, high voluntary turnover is a serious concern. Each time an employee leaves, the company not only loses a skilled worker right away but also loses important knowledge and experience that person had about how things work in the organization. Furthermore, replacing a departing employee can be very costly, often ranging between 30 percent and 400 percent of the person's annual salary. These costs include recruitment, onboarding, training, and temporary losses in productivity. In the context of route expansion and new operational plans, retaining experienced staff becomes even more critical to ensure service continuity and quality.

The causes behind such turnover can be better understood using the **Organization Equilibrium Theory**, which explains that employees decide to leave when they believe their contributions to the company outweigh the incentives they receive in return. If employees at FAU Transline feel that their effort, time, or emotional labor are not matched by fair compensation, recognition, or career development, they may begin to consider other job options. This perceived imbalance directly affects job satisfaction and can lead to turnover intentions. Additionally, if external opportunities are readily available and barriers to change are low, the likelihood of an employee resigning becomes higher.

2.2 Employee Turnover Theories (6 Points)

Case 1: Jonas

The most appropriate explanation for Jonas's decision to leave FAU Transline can be found in the **Unfolding Model of Turnover**, particularly **Path 2**. This model highlights how employee exits are often triggered by a "shock," which is any notable event that leads an employee to re-evaluate their alignment with the organization. In Jonas's case, the organizational restructuring, involving a shift from a decentralized to a centralized structure, serves as the shock. This shock did not align with his preferred way of working, which was characterized by autonomy, informality, and close interpersonal relations.

Because this shock did not activate any pre-existing script or plan to quit immediately, Jonas likely went through a cognitive evaluation process. He experienced what is termed an **image violation**, where the new organizational structure conflicted with his values and work identity. Jonas valued independence and informal work settings, but the new standardized, hierarchical structure clashed with these preferences. This mismatch led to dissatisfaction, and even without an alternative offer, he chose to leave. This matches the decision-making flow

described in Path 2, where a shock leads to image violation and dissatisfaction but no search for alternatives or offers is necessarily present.

Case 2: Elena

Elena's situation is best explained by the **Job Embeddedness Theory**, which focuses on why employees choose to stay or leave based on how embedded they are within their work and community environments. While Elena enjoys the purpose of her work and aligns with the company's values around customer service, she lacks **social connection** within the organization. According to the theory, employees are more likely to stay when they feel linked to colleagues and experience a strong sense of fit. Elena's ongoing sense of isolation and her inability to build meaningful workplace relationships reduce her level of embeddedness. If someone doesn't have strong social connections at work, it's emotionally easier for them to leave, so quitting becomes a more likely option.

To improve employee retention in cases like Elena's, FAU Transline can focus on enhancing **social support and community integration**. The company could introduce structured onboarding programs, mentorship initiatives, or team-building activities that actively foster relationships among colleagues. Managers should be trained to identify signs of social disconnection early and take supportive actions. Strengthening these workplace connections not only increases employee well-being but also raises the likelihood that employees will feel more embedded and thus remain with the organization.

2.3 Employee Turnover Dataset (10 Points)

The dataset from FAU Transline includes detailed employee records across various job roles such as bus drivers, train conductors, and maintenance technicians. It contains information relevant to employee turnover including salary levels, job satisfaction rates, duration at the company, and promotion history. The objective was to analyze turnover trends and preprocess the dataset for further analysis.

Data Import and Loading

The essential Python libraries for data handling and visualization were imported. The turnover.csv dataset, containing employee records of FAU Transline, was loaded into a pandas DataFrame for analysis.

Analysis of Salary Variation Within Each Job Role

To examine how salary varies across different job roles at FAU Transline, a pivot table was created to count the number of employees within each salary category per job role. This table summarized the distribution of low, medium, and high salary levels for each job role. A stacked bar chart was then plotted using this pivot table, visually representing the variation in salary levels within each role. The chart clearly illustrated the proportion and count of employees earning different salary levels across all job roles.



Average Job Satisfaction of Employees Who Left

To understand the job satisfaction level of employees who voluntarily left FAU Transline, the dataset was filtered to include only those with a “left” status of 1. The average job satisfaction rate among these employees was then calculated.

2. Average job satisfaction level of employees who left

```
employees_who_resigned = df[df['left'] == 1]
avg_job_satisfaction_level = employees_who_resigned['job_satisfaction_rate'].mean()
print(f"Average job satisfaction of employees who left: {avg_job_satisfaction_level:.2f}")
```

Average job satisfaction of employees who left: 0.44

Average Duration of Stay for Employees Who Left

Similarly, employees who left were filtered from the dataset. Then, the mean duration of their stay at the company was calculated.

3. Average duration employees who left stayed with the company

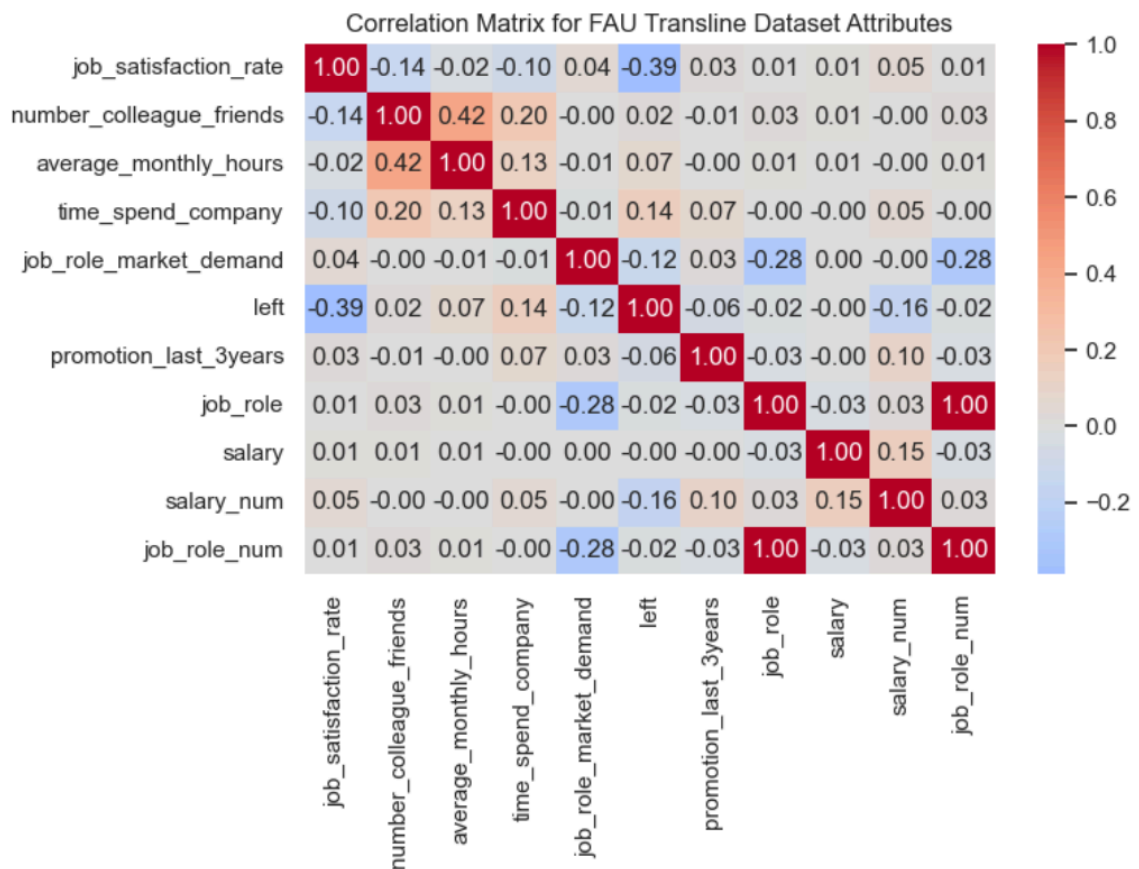
```
avg_duration_time_spent = employees_who_resigned['time_spend_company'].mean()
print(f"Average duration (years) of stay for employees who resigned: {avg_duration_time_spent:.2f} years")
```

Average duration (years) of stay for employees who resigned: 3.88 years

Correlation Analysis with Employee Turnover

The correlation matrix of the dataset was computed and visualized using a heatmap. The correlations of all features with the **left** variable were extracted, rounded, and sorted. Findings show that **job_satisfaction_rate** has the strongest negative correlation (-0.39) with employee turnover, indicating higher satisfaction reduces leaving. **time_spend_company** (0.14) and

average_monthly_hours (0.07) show weak positive correlations. Salary-related features exhibit negligible or negative correlations, suggesting limited impact on turnover in this dataset.



Data Preprocessing

The dataset contained categorical variables that needed conversion to numeric format for analysis.

- The **job_role** column, originally a string type representing different job roles, was transformed into numeric labels using a label encoding technique. This assigns a unique integer to each distinct job role.
- Similarly, the **salary** column, which contained categorical salary levels, was converted to numeric values using label encoding.

Both transformations were done in place on the original DataFrame to prepare the data for further modeling and analysis. This step ensures that machine learning algorithms can interpret the categorical information correctly.

2.4 Turnover prediction (5 Points)

Choice of Machine Learning Model

For predicting employee turnover at FAU Transline, the Random Forest classifier was selected due to its strong ability to handle complex, nonlinear relationships among features without requiring extensive parameter tuning. It is an ensemble method that combines multiple decision trees to reduce overfitting and improve generalization. The dataset contains a mixture of numeric and encoded categorical features, which Random Forest handles effectively. Furthermore, it provides insights into feature importance, which is valuable for understanding key factors influencing employee turnover.

Model Performance Evaluation

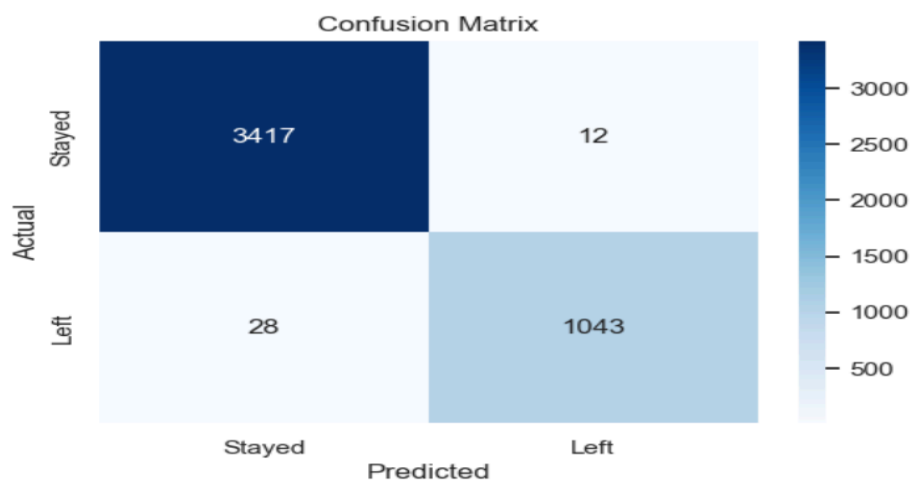
The Random Forest model achieved solid predictive performance with an accuracy of approximately 99%, along with good precision and recall values, indicating reliable identification of employees who leave the organization. The classification report and confusion matrix confirm that the model balances false positives and false negatives reasonably well. These metrics demonstrate that the model can be a useful tool for FAU Transline to anticipate which employees might be at risk of leaving.

Model Performance on Test Set:

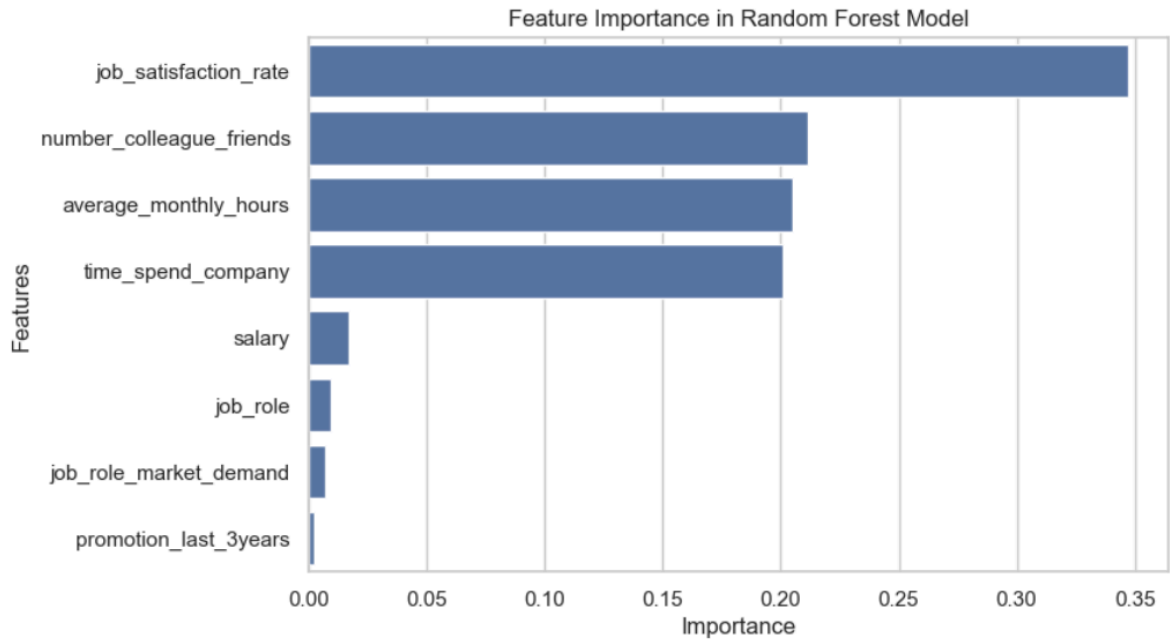
Accuracy: 0.99
Precision: 0.99
Recall: 0.97
F1 Score: 0.98

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	3429
1	0.99	0.97	0.98	1071
accuracy			0.99	4500
macro avg	0.99	0.99	0.99	4500
weighted avg	0.99	0.99	0.99	4500



Indications for Employee Turnover at FAU Transline



The feature importances suggest employees are mainly leaving due to **low job satisfaction**, lack of **colleague friendships**, **long working hours** and **time spent at the company**. Salary and promotions have a minor impact here compared to other factors.

Recommendation to retain employee

To retain employees, FAU Transline should:

- Improve job satisfaction by addressing work conditions and employee engagement.
- Foster a friendly, collaborative work environment to build strong colleague relationships.
- Manage workloads to prevent burnout from excessive hours.
- Consider competitive salary reviews and promotion opportunities, even if their current impact is smaller.

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